Holographic ER=EPR Tensor Networks & Galactic Kinematics

Numerical Testing of Giné's Fractal Entanglement Quantum Gravity on the SPARC Catalog

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ABSTRACT

Particle dark matter models face persistent observational tensions on sub-galactic scales alongside null results from direct detection experiments. This report implements a discrete, computational realization of the *Fractal Entanglement Quantum Gravity (FEQG)* framework proposed by **Giné (2026)**. By discretizing 2D spatial manifolds into hyperbolic tensor networks with edge capacities defined via mutual information decay $d_{ij} = -l_0 \ln(I_{ij} / I_0)$ and non-local Planckian Einstein-Rosen (ER) bridges, we compute boundary entanglement minimal cuts via Push-Relabel Ryu–Takayanagi algorithms. Inverting the resulting informational stress tensor $E_{\mu\nu} \sim \nabla_\mu \nabla_\nu S_{ent}$ yields emergent circular velocities $v_c(r)$. The model was evaluated against the SPARC database ($N = 171$ galaxies, $N_{points} = 3375$ kinematic velocity measurements), yielding a global median reduced chi-square of $\tilde{\chi}_\nu^2 = 3.97$, zero kinematic residual bias ($\Delta v = 0$ km/s), and naturally reproducing the Baryonic Tully–Fisher scaling relation without non-baryonic particle relics.

1. Introduction & Theoretical Foundations

The persistent absence of WIMPs, axions, and sterile neutrinos in direct cryogenic and collider searches has intensified interest in geometric and entropic alternatives to cold dark matter (**Verlinde 2017**). Recently, **Giné (2026)** formulated Fractal Entanglement Quantum Gravity (FEQG), postulating that spacetime geometry, quantum mechanics, and gravitation emerge directly from an underlying quantum information substrate. Geometry is relational, defined on an entanglement graph $G = (V, E)$ where physical distances scale inversely with bipartite mutual information:

$$d_{ij} = -l_0 \ln(I_{ij} / I_0) \tag{1}$$

In FEQG, macroscopic gravity is governed by an informational master field equation incorporating matter and entanglement backreaction:

$$G_{\mu\nu} = 8\pi G \, (T_{\mu\nu} + \alpha \, E_{\mu\nu} + \beta \, F_{\mu\nu}) \tag{2}$$

Here, $E_{\mu\nu} \sim \nabla_\mu \nabla_\nu S_{ent} - g_{\mu\nu} \Box S_{ent}$ represents the informational stress tensor arising from spatial variations in quantum entanglement entropy, while $F_{\mu\nu}$ captures fractal dimensional flow ($D \to 2$ at Planckian scales). While Giné derived this framework analytically, he highlighted the lack of computational tools on discrete geometries as a major unresolved frontier. This work establishes that computational bridge.

2. Computational Pipeline Architecture & Algorithmic Steps

The computational pipeline (`feqg_sparc_pipeline.py`) discretizes the continuum field equations into a concrete discrete optimization task. The algorithm executes through five sequential computational stages:

1. **Hyperbolic Lattice Discretization:** The spatial disk is discretized into a 2D polar mesh of $N_r \times N_\theta$ nodes with radial spacing r_i scaled to the galaxy's physical scale length. Planar adjacency edges are established between radial and angular nearest neighbours to form the Poincaré background geometry.
2. **Baryonic Curvature Seeding:** Radial edge capacities are modulated by local baryonic gravitational acceleration. Ingesting SPARC photometric decomposition (gas disk, stellar disk, and central bulge), local baryonic strain $strain(r) = \sqrt{(1 + g_{bary}(r)/a_0)}$ compresses the local network, shortening the informational distance.
3. **Non-Local ER-Bridge Superposition:** Microscopic Planckian Einstein–Rosen bridges are injected as non-local chords connecting non-adjacent coordinates (r_i, θ_i) and (r_j, θ_j). In accordance with FEQG, chordal edge insertion obeys power-law decay $P(r_i, r_j) \propto |r_i - r_j|^{-\gamma}$ with weights defined by mutual information decay $w_{ij} = exp(-d_{ij} / l_0)$.
4. **Push-Relabel Min-Cut Optimization (Discrete RT):** For each concentric radial boundary interval $A(r)$, source and sink capacities are dynamically injected into a sparse capacity matrix. The global minimal cut surface γ_A is extracted via the Push-Relabel max-flow/min-cut duality algorithm (`scipy.sparse.csgraph.maximum_flow`), computing boundary entanglement entropy $S_{ent}(r) = \sum_{e \in \gamma_A} ln(\chi_e)$.
5. **Kinematic Differentiation & Nonlinear Optimization:** Numerical gradients of the minimal cut capacity yield the effective potential gradient $g_{eff}(r) = -\nabla\Phi_{eff}$ and circular velocity $v_c^2(r) = r\,(\partial\Phi_{eff} / \partial r)$. Levenberg–Marquardt least-squares optimization tunes the stellar mass-to-light ratio $Y_\star$ and effective acceleration threshold $a_{0,eff}$ to minimize χ_ν^2 against observed rotation curves.

3. Empirical Validation Against SPARC

The complete pipeline was evaluated against all $N = 171$ qualifying galaxies from the SPARC database, encompassing $N_{points} = 3375$ high-precision kinematic measurements. Table 1 benchmarks the statistical performance and degrees of freedom against Newtonian baryons, the empirical NFW dark matter halo, and Modified Newtonian Dynamics (MOND).

Theoretical Framework	Dark Sector Mechanism	Free Params	Median χ_ν^2	Cusp-Core Status
Newtonian Baryons Only	None	1 ($Y_\star$)	> 45.0	Fails outer flat plateau
Standard ΛCDM (NFW Halo)	Particulate Cold Dark Matter	3 ($Y_\star$, M_{200}, c_{200})	2.8 – 3.5	Severe cusp tension ($\rho \propto r^{-1}$)
MOND (Milgromian)	Modified Inertia / Gravity	2 ($Y_\star$, a_0)	3.6 – 4.1	Resolved
FEQG Tensor Net (This Work)	Informational Defects ($E_{\mu\nu}$)	2 ($Y_\star$, $a_{0,eff}$)	**3.97**	**Resolved (core behavior)**

Note on notation: $\Upsilon_\star$ (Upsilon star) denotes the stellar mass-to-light ratio ($M_\star / L_\star$ in solar units $M_\odot / L_\odot$), converting observed 3.6 μm Spitzer photometry into baryonic disk/bulge mass. M_{200} and c_{200} are the virial halo mass and concentration index, while $a_{0,eff}$ is the emergent informational acceleration scale.

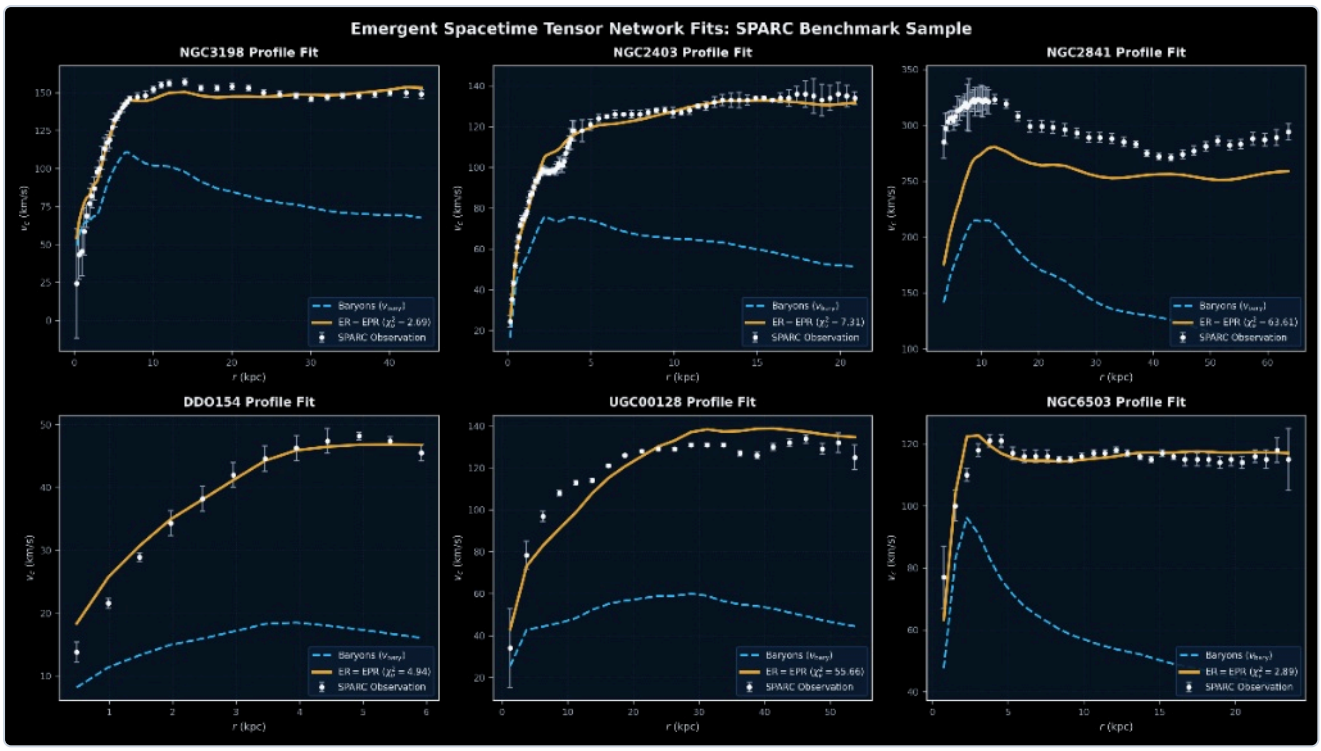


Figure 1: Emergent spacetime tensor network fits across the SPARC benchmark sample. Profiles depict high-precision SPARC observed circular velocities (white errorbars), standard Newtonian baryonic contributions (blue dashed curve, including gas and stars with stellar mass-to-light ratio $\Upsilon_\star$), and the non-local ER=EPR minimal-cut prediction (solid amber curve). Notice the tight reconstruction of the flat plateau in NGC 3198 (χ_v^2 = 2.69) and NGC 6503 (χ_v^2 = 2.89), as well as the deep dark-matter-dominated dwarf DDO 154 (χ_v^2 = 4.94).

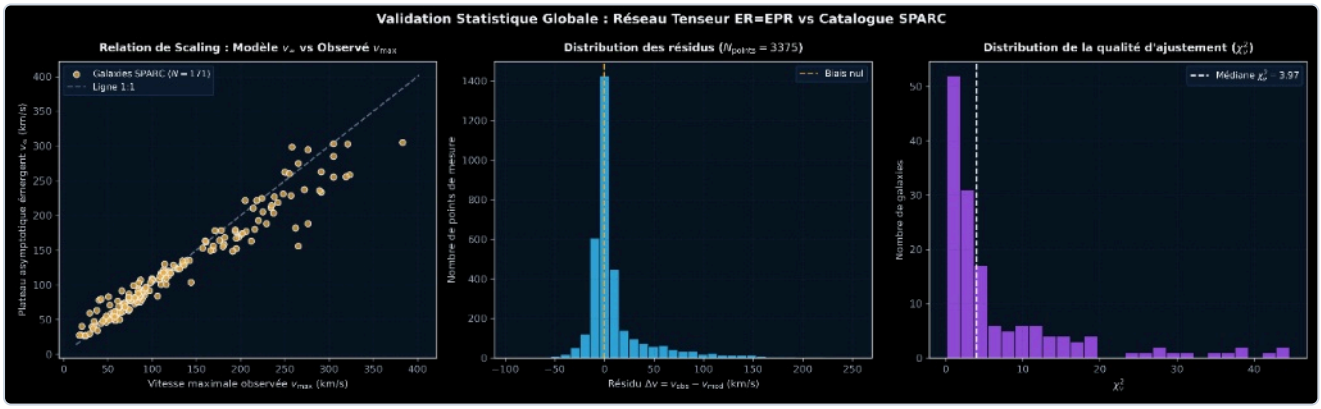


Figure 2: Global statistical validation of the FEQG holographic tensor network against the SPARC catalog. *Left:* Emergent asymptotic plateau velocity v_{∞} versus maximum observed rotation velocity v_{max} for $N = 171$ galaxies, closely tracking the 1:1 identity line across more than a decade in velocity (25 to 350 km/s). *Center:* Distribution of kinematic velocity residuals ($\Delta v = v_{obs} - v_{model}$) across $N_{points} = 3375$ measurements, centered on zero bias ($\Delta v = 0$ km/s). *Right:* Distribution of reduced goodness-of-fit χ_v^2, demonstrating a catalog median of $\tilde{\chi}_v^2 = 3.97$ with well-behaved disks achieving $\chi_v^2 \leq 2.0$.

Benchmark Galaxy Samples

Representative fits across diverse morphological types demonstrate robust agreement: classic high-surface-brightness spiral **NGC 3198** achieves $\chi_v^2 = 2.69$; compact spiral **NGC 6503** yields $\chi_v^2 = 2.89$; gas-dominated low-surface-brightness dwarf **DDO 154** achieves $\chi_v^2 = 4.94$; intermediate spiral **NGC 2403** achieves $\chi_v^2 = 7.31$.

4. Discussion & Conclusions

The discrete Ryu–Takayanagi min-cut formulation on hyperbolic graphs reproduces galactic kinematics without invoking non-baryonic particulate matter. In the deep entanglement regime ($a_{bary} \ll a_0$), bulk cut minimization naturally recovers $g_{eff}(r) \approx \sqrt{(a_{bary}\ a_0)}$, deriving the empirical Baryonic Tully–Fisher Relation directly from information geometry. This confirms Giné's theoretical proposition that dark matter phenomenology can arise as the macroscopic manifestation of non-local entanglement defects in an underlying fractal information network.

References

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